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### Simulated candle flame structure of electronic candle

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#### Abstract

A simulated candle flame structure of an electronic candle is provided. The electronic candle includes a candle body. A flame simulating board is mounted on a top end of the candle body. A light emitting element is arranged in the interior of the candle body. The light emitting element emits light toward the flame simulating board to be reflected thereby so as to simulate the appearance of a candle in burning. A wick flame pattern is provided on a bottom end of the flame simulating board. The wick flame pattern can be directly transfer-printed on the flame simulating board or is adhesively attached to the flame simulating board, so that the light from the light emitting element is reflected to demonstrate the appearance of a wick burning in a flame, thereby displaying a more realistic electronic candle.

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## Background/Summary

### BACKGROUND OF THE INVENTION

#### (a) Technical Field of the Invention

(1) The present invention relates generally to an electronic candle, and more particularly to an electronic candle, which produces a visual sense of a candle wick in a simulated candle flame after being lit, so that the electronic candle has a more realistic effect in use.

#### (b) Description of the Prior Art

(2) Candles produce lighting effects by burning their wicks, but because they use an open flame, they may ignite surrounding objects and cause accidents such as fires. For this reason, in recent years, electronic candles have been proposed, which use the projection of light to simulate the burning state of a candle. The electronic candles are reusable and do not produce smoke, and occurrence of accidents such as fires can be avoided because they do not use an open flame for burning.

(3) An existing electronic candle, as shown in Taiwan Utility Model M474857 “Electronic Candle”, comprises a hollow shell, a movable reflection plate assembly, a circuit board, and a light-emitting diode (LED) light. The hollow shell has an upper opening and a bottom surface opposite to the

upper opening. The movable reflection plate assembly comprises a suspension rod and a reflection plate, wherein two ends of the suspension rod are connected to inner walls of the hollow shell. The reflection plate is suspended from the suspension rod and has a reflective end protruding out of the upper opening and a magnetic induction end opposite to the reflective end. A coil is provided on the circuit board, and the coil generates a time-varying magnetic field. The LED light is electrically connected to the circuit board, and a light-emitting surface of the LED light faces the reflective end. When the electronic candle of this Utility Model is operating, the magnetic induction end will be pushed by the time-varying magnetic field, causing the reflection plate to sway to thereby simulate a swaying effect of a candle.

(4) The reflection plate disclosed in the above-mentioned prior art is a pure white reflection plate, which is capable of reflecting light from the LED light and producing a flame effect, yet when the candle burns, the bottom of the flame is burning upward from the periphery of a wick. The existing electronic candle operates by simply reflecting the light from the LED light and cannot simulate the effect of the flame burning upward from the candle wick. Therefore, it cannot simulate the burning state of a real candle, and further improvement is necessary.

(5) The inventor of the present invention performed research and innovation to provide the simulated candle flame structure of the electronic candle described in the present invention.

#### SUMMARY OF THE INVENTION

(6) An objective of the present invention is to provide a simulated candle flame structure of an electronic candle, wherein the electronic candle comprises a candle body, a flame simulating board being mounted on a top end of the candle body, a light emitting element being arranged in an interior of the candle body, wherein the light emitting element is operable to emit light toward the flame simulating board to be reflected thereby for simulating an appearance of a candle burning, wherein the flame simulating board has a board surface on which a wick flame pattern is formed, the wick flame pattern being directly printed or painted on the flame simulating board or adhesively attached to the flame simulating board.

(7) In the simulated candle flame structure of the electronic candle disclosed in the present, the flame simulating board comprises a flame color gradient area on a periphery thereof, and the flame color gradient area gradually becomes lighter and brighter from the periphery toward a center.

(8) In the simulated candle flame structure of the electronic candle disclosed in the present, the flame color gradient area and the wick flame pattern are printed on a sticker that is pasted on the flame simulating board.

(9) In the simulated candle flame structure of the electronic candle disclosed in the present, the flame color gradient area and the wick flame pattern are directly printed or painted on the flame simulating board.

(10) In the simulated candle flame structure of the electronic candle disclosed in the present, a color of the wick flame pattern gradually becomes lighter and brighter from bottom to top.

(11) In the simulated candle flame structure of the electronic candle disclosed in the present, the flame simulating board comprises a plate having a curved surface or a flat plate, the present invention is not limited thereto.

(12) The desirable entities disclosed in the present invention can be clearly understood from the following description and drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic exploded view showing the present invention adhesively attached to a flame simulating board.

(2) FIG. 2 is a schematic perspective view of FIG. 1 in an assembled form.

(3) FIG. 3 is a partial enlarged view of FIG. 2.

(4) FIG. 4 is a schematic view showing a wick flame pattern and a flame color gradient area printed on a sticker according to the present invention.

(5) FIG. 5 is a schematic view showing a use of a known electronic candle.

(6) FIG. 6 is a schematic view showing a use of the present invention.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) As shown in FIGS. 1-4, the present invention relates to a simulated candle flame structure of an electronic candle. The electronic candle comprises a candle body **10**. A flame simulating board **11** is mounted on a top end of the candle body **10**, and a light emitting element **12** is mounted in an interior of the candle body **10**. The light emitting element **12** emits light projecting toward the flame simulating board **11** to be reflected thereby so as to simulate the appearance of a candle burning. The above-described structure is known in the prior art and has been commonly used in the structures of electronic candles, such as that shown in Taiwan Utility Model M474957, and is not a novel part of the present invention, and no further details will be described herein.

(8) The simulated candle flame structure of the electronic candle according to the present invention features that a wick flame pattern **20** is provided on a board surface of the flame simulating board **11**. The wick flame pattern **20** can be directly printed or coated on the flame simulating board **11** or can be printed on a sticker that is attached on the flame simulating board **11**. The present invention does not impose any constraint to how the wick flame pattern **20** is arranged on the flame simulating board **11**.

(9) In the simulated candle flame structure of the electronic candle according to the present invention, the flame simulating board **11** further comprises a flame color gradient area **30** arranged on a periphery thereof. The flame color gradient area **30** gradually becomes lighter and brighter from the periphery of the flame simulating board **11** toward the center. The flame color gradient area **30** can exhibit a gradient color from red to white or bright yellow towards the center, or a gradient color from orange red to white or bright yellow towards the center, but the present invention is not limited thereto.

(10) In the simulated candle flame structure of the electronic candle according to the present invention, the flame color gradient area **30** and the wick flame pattern **20** can be printed on a sticker and pasted on the flame simulating board **11**, as shown in FIG. 4; or alternatively, the flame color gradient area **30** and the wick flame pattern **20** are directly printed or painted on the flame simulating board **11**, and the present invention is not limited thereto.

(11) As shown in FIGS. 3 and 4, in the simulated candle flame structure of the electronic candle according to the present invention, the color of the wick flame pattern **20** gradually becomes lighter and brighter from the bottom to the top. The wick flame pattern **20** can be black or brown and gradually fade upward to white or bright yellow to present a realistic appearance of a burning candle wick.

(12) In the simulated candle flame structure of the electronic candle according to the present invention, the flame simulating board **11** can be a curved plate, which focuses light toward the center, or the flame simulating board **11** can be a flat plate, and the present invention is not limited thereto.

(13) As shown in FIGS. 2 and 3, when light emitting from a light source toward the flame simulating board **11** described in the present invention, the wick flame pattern **20** is shown and highlighted. Further, the flame color gradient area **30** on the periphery of the wick flame pattern **20** also helps make the periphery darker than the center, thus producing the appearance of a real candle in burning. Referring to FIG. 5, which is a photo showing a use of an existing electronic candle, the existing electronic candle uses pure white color to form a flame simulating board, which shows only flame without exhibiting a “wick”. Oppositely, referring to FIGS. 4 and 6, wherein FIG. 6 is a photo showing a use of the present invention, when the present invention projects light to the flame simulating board, the light highlights the wick flame pattern **20**, and the flame color gradient area

**30** on the periphery of the wick flame pattern **20** also makes the periphery darker than the center of the flame simulating board, thereby producing the appearance of a real candle in burning, so that the electronic candle according to the present invention can better simulate the lighting state of a real candle, thereby demonstrating the novelty and excellence of the present invention.

## Claims

1. A simulated candle flame structure of an electronic candle, the electronic candle comprising a candle body, a flame simulating board being mounted on a top end of the candle body, a light emitting element being arranged in an interior of the candle body, wherein the light emitting element is operable to emit light toward the flame simulating board to be reflected thereby for simulating an appearance of a candle burning, wherein the flame simulating board has a board surface on which a wick flame pattern is formed, the wick flame pattern being directly printed or painted on the flame simulating board or adhesively attached to the flame simulating board; wherein the flame simulating board comprises a flame color gradient area on a periphery thereof, the flame color gradient area gradually becomes lighter and brighter from the periphery toward a center, and a color of the wick flame pattern gradually becomes lighter and brighter from bottom to top.
  2. The simulated candle flame structure of the electronic candle according to claim 1, wherein the flame color gradient area and the wick flame pattern are printed on a sticker that is pasted on the flame simulating board.
  3. The simulated candle flame structure of the electronic candle according to claim 1, wherein the flame color gradient area and the wick flame pattern are directly printed or painted on the flame simulating board.
  4. The simulated candle flame structure of the electronic candle according to claim 1, wherein the flame simulating board comprises a plate having a curved surface.
  5. The simulated candle flame structure of the electronic candle according to claim 1, wherein the flame simulating board comprises a flat plate.
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